

SET 11: PRODUCT CATEGORY ANALYSIS

Dataset: Product Categories

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Dataset: Product Categories

Product Category	Sales (in \$)
Electronics	50000
Clothing	35000
Appliances	40000

PROGRAM 1: Funnel Chart of Sales by Product Category

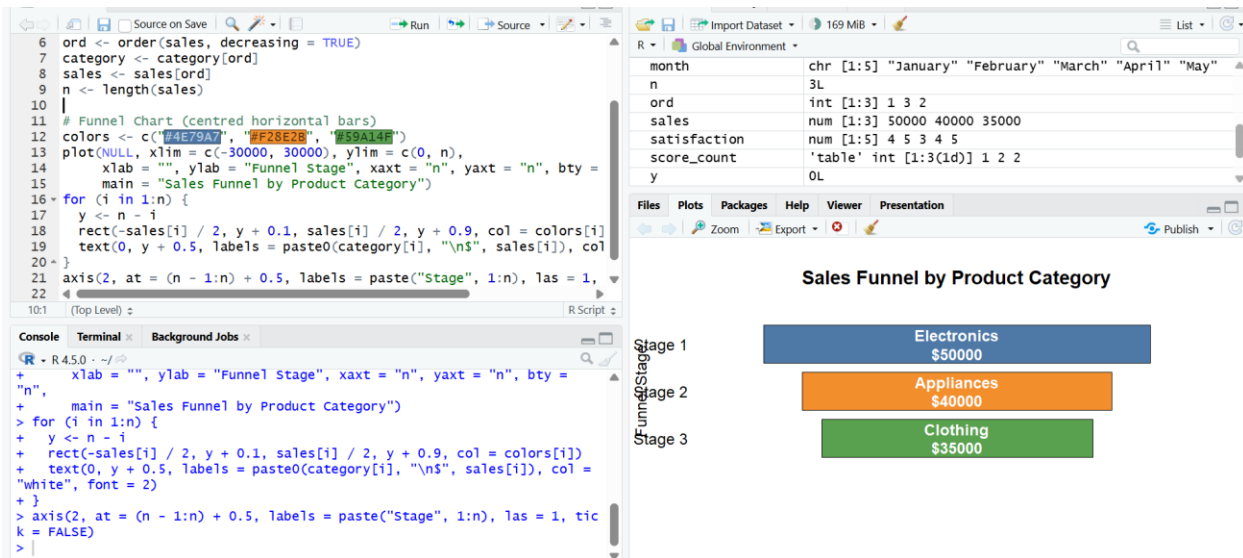
Question: Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

```
# Dataset
category <- c("Electronics", "Clothing", "Appliances")
sales <- c(50000, 35000, 40000)

# Sort from largest to smallest so the funnel narrows at each stage
ord <- order(sales, decreasing = TRUE)
category <- category[ord]
sales <- sales[ord]
n <- length(sales)

# Funnel Chart (centred horizontal bars)
colors <- c("#4E79A7", "#F28E2B", "#59A14F")
plot(NULL, xlim = c(-30000, 30000), ylim = c(0, n),
      xlab = "", ylab = "Funnel Stage", xaxt = "n", yaxt = "n", bty = "n",
      main = "Sales Funnel by Product Category")
for (i in 1:n) {
  y <- n - i
  rect(-sales[i] / 2, y + 0.1, sales[i] / 2, y + 0.9, col = colors[i])
  text(0, y + 0.5, labels = paste0(category[i], "\n$", sales[i]), col =
"white", font = 2)
}
axis(2, at = (n - 1:n) + 0.5, labels = paste("Stage", 1:n), las = 1, tick =
FALSE)
```

Output 1: Sales Funnel Chart



Observation: The funnel narrows from Electronics (\$50,000) at Stage 1 to Appliances (\$40,000) at Stage 2 and Clothing (\$35,000) at Stage 3, showing Electronics as the strongest category and Clothing as the weakest.

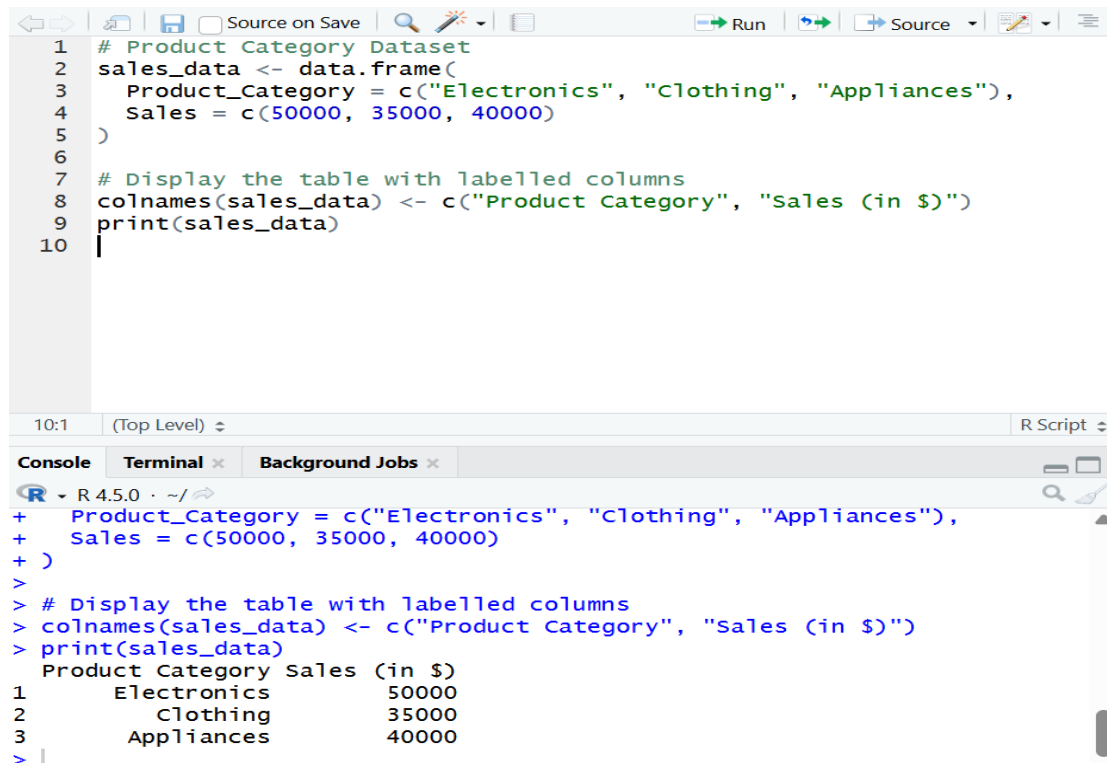
PROGRAM 2: Product Category Sales Table

Question: Build a table to display the sales data for each product category. Label the table elements.

```
# Product Category Dataset
sales_data <- data.frame(
  Product_Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)

# Display the table with labelled columns
colnames(sales_data) <- c("Product Category", "Sales (in $)")
print(sales_data)
```

Output 2: Product Category Sales Table



```
1 # Product Category Dataset
2 sales_data <- data.frame(
3   Product_Category = c("Electronics", "Clothing", "Appliances"),
4   Sales = c(50000, 35000, 40000)
5 )
6
7 # Display the table with labelled columns
8 colnames(sales_data) <- c("Product Category", "Sales (in $)")
9 print(sales_data)
10
```

Console

```
+ Product_Category = c("Electronics", "Clothing", "Appliances"),
+ Sales = c(50000, 35000, 40000)
+ )
>
> # Display the table with labelled columns
> colnames(sales_data) <- c("Product Category", "Sales (in $)")
> print(sales_data)
  Product Category Sales (in $)
1    Electronics    50000
2     Clothing    35000
3    Appliances    40000
>
```

Observation: The table displays the Product Category and Sales (in \$) for each of the three categories. Total sales across all categories are \$125,000.

PROGRAM 3: Interactive Tableau Dashboard

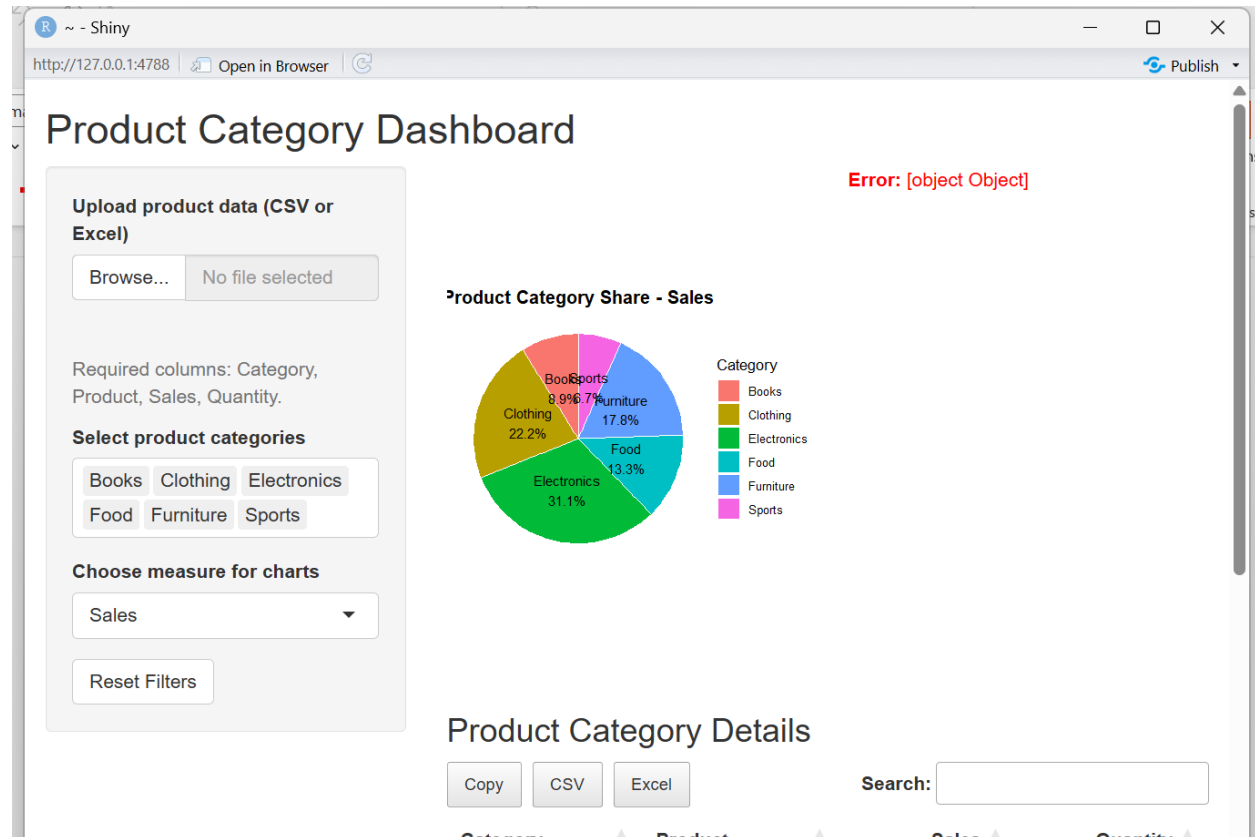
Question: Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.

Tableau Steps

1. Open Tableau and connect to the Product Categories dataset (Excel/CSV).
2. Verify the fields Product Category and Sales (in \$).
3. Create a worksheet named "Pie Chart". Drag Product Category to Color and Label, and Sales to Angle and Label; select Pie in the Marks card.
4. Add the title "Sales Distribution by Product Category" and show percentages (Quick Table Calculation > Percent of Total) in the labels.
5. Create a worksheet named "Funnel Chart". Drag Sales to Columns and Product Category to Rows, then sort the categories in descending order of Sales.
6. Select Bar in the Marks card, drag Sales to Size, and set the bars to be centred: create a calculated field "Neg Sales" = $-\text{SUM}([\text{Sales}])/2$, place both it and $\text{SUM}([\text{Sales}])/2$ on Columns as a dual axis, and synchronise the axes so the bars form a funnel.
7. Add the title "Sales Funnel by Product Category", label each stage with Product Category and Sales, and hide the axis.
8. Create a worksheet named "Sales Table". Drag Product Category to Rows and Sales to Text.

9. Create a new Dashboard, set the size to Automatic, and drag in the Pie Chart, Funnel Chart and Sales Table worksheets.
10. Add the title “Product Category Dashboard”.
11. Select the pie chart and click the “Use as Filter” icon; add a Product Category quick filter (Show Filter) so selections update all views.
12. Test the dashboard by clicking a pie slice or choosing a category in the filter and checking that the funnel chart and table update.

Output 3: Product Category Dashboard



Observation: The Tableau dashboard combines the pie chart, funnel chart and sales table, allowing interactive exploration of product category sales through filters and click-to-highlight actions.

PROGRAM 4: Pie Chart of Sales Distribution

Question: Create a pie chart to represent the distribution of sales across product categories. Include labels.

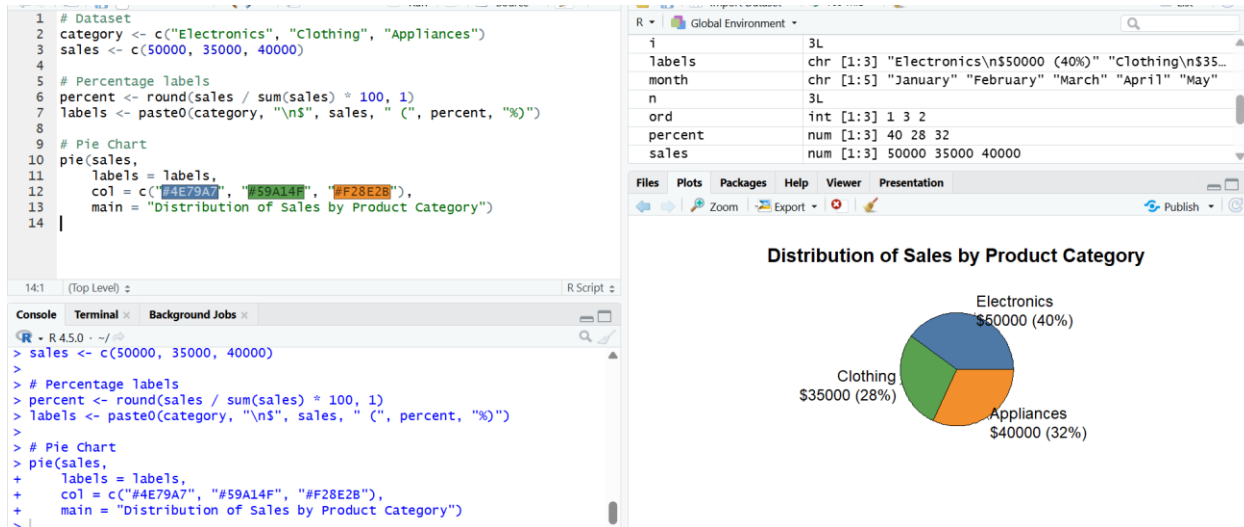
```
# Dataset
category <- c("Electronics", "Clothing", "Appliances")
sales <- c(50000, 35000, 40000)

# Percentage labels
percent <- round(sales / sum(sales) * 100, 1)
labels <- paste0(category, "\n$", sales, " (", percent, "%)")

# Pie Chart
```

```
pie(sales,
    labels = labels,
    col = c("#4E79A7", "#59A14F", "#F28E2B"),
    main = "Distribution of Sales by Product Category")
```

Output 4: Sales Distribution Pie Chart



Observation: Electronics contributes the largest share of sales at 40%, followed by Appliances at 32% and Clothing at 28%.

Procedure

1. Create the product category sales data in R.
2. Sort the sales values in descending order and draw the funnel chart using `rect()`, labelling each stage and adding a title.
3. Create a data frame containing Product Category and Sales, label the columns and display it as a table.
4. Open Tableau and import the Product Categories dataset.
5. Create the pie chart, funnel chart and sales table as separate worksheets.
6. Combine the worksheets into a dashboard, add a title, and enable filters.
7. Test the dashboard interactivity.
8. Calculate the percentage share of each category and draw the labelled pie chart using `pie()`.

Result

The Product Category sales data was successfully visualized using a funnel chart, a labelled sales table, an interactive Tableau dashboard, and a pie chart showing the distribution of sales.